Anmol Goyal

anmol.goyal@kcl.ac.uk · +44 7778 701965 · linkedin.com/in/anmolxgoyal · London, UK

Education

BSc (Hons) Mathematics with Statistics *Sep 2024 – Jul 2027*
King's College London *London, UK*

- **Final-year modules:** Informatics modules in Natural Language Processing and Computational Modelling, alongside core mathematics and statistics.
- **Prior modules:** Probability & Statistics II, Applied Probabilistic Models, Random Structures & Algorithms, Statistical Modelling, Real Analysis, Discrete Mathematics.

Experience

Oxford Dynamics *(defence AI & robotics)* *Harwell Science Campus, Oxfordshire*
Visiting Student AI Researcher *Sep 2026 – Present*

- Continuing with the company after the internship to lead research on **liquid time-constant (LTC) networks as adaptive filters**, run as my KCL final-year project under joint academic and industry supervision.
- Investigating whether the learned, input-dependent time constants of LTC networks correspond to the Kalman gain of an optimal linear filter, connecting continuous-time neural ODEs with classical state estimation; built on open-source datasets so the work remains publishable.

Robotics Intern *Jun 2026 – Sep 2026*

- Wrote ROS 2 / MoveIt motion-control scripts for a 6-DoF Universal Robots UR7e collaborative arm; this code runs on the hardware and is merged into the team repository.
- Contributed to commissioning of the UR7e and built a digital twin of the arm in NVIDIA Isaac Sim for simulation-first testing of motion plans.
- Developed ROS 2 nodes and publisher–subscriber pipelines in containerised Docker environments, working to the team's branching and code-review practices.
- Conducted literature reviews on AI foundations and world models to inform the company's research direction for learned models in autonomous systems.
- Reviewed the open literature on classical guidance and control, including null-seeking guidance, to ground modelling work in established control theory.

Technical Skills

Languages: Python (7 years, self-taught), SQL, R, JavaScript, Java, LaTeX
Robotics: ROS 2, MoveIt, NVIDIA Isaac Sim, Universal Robots (UR7e)
Tooling: Git, Docker, Linux
Data & ML: pandas, matplotlib, BeautifulSoup (web scraping), statistical modelling (regression, diagnostics, model selection)
Mathematics: Probability, stochastic processes, state estimation (Kalman filtering), linear algebra

Leadership & Outreach

Co-President, IBM Z Society & IBM Z Student Ambassador, KCL – hosted campus talks with Amazon engineers and an IBM director *Jun 2025 – Present*
NMES Faculty Student Representative, KCL – analysed ∼150 survey responses and presented findings to academic staff *Oct 2025 – Present*
Subject Ambassador, KCL Mathematics – explained technical topics to audiences of 20–60 at outreach events *Nov 2025 – Present*
Volunteer Python Instructor, Code Your Future – taught beginners in groups of 4–6 *Aug 2024 – Jan 2025*

Languages: English (fluent), Hindi (native), Punjabi (intermediate) · **Certifications:** IBM Z Xplore Concepts, IBM Z & LinuxONE Community Contributor